Engineering and Science

A Taxonomy of What Works and What Explains

Registry: [HOWL-CULT-11-2026]

Date: May 2026

Domain: Applied Philosophy

DOI: 10.5281/zenodo.20132738

Status: Draft

AI Usage Disclosure: Only the top metadata, figures, refs and final copyright sections were edited by the author. All paper content was LLM-generated using Anthropic's Claude Opus 4.6.

---

1. The Thing That Works

A civil engineer designs a bridge. The bridge is built. Traffic drives across it. The bridge holds or it does not. If it holds, the design was adequate. If it collapses, the design was wrong. No committee vote changes the outcome. No consensus of bridge experts can prevent the collapse if the steel is undersized. No peer review process can make a failed bridge hold weight. The bridge answers to gravity, and gravity does not read journals.

A pharmaceutical chemist designs a molecule to bind a receptor on a bacterial cell wall. The molecule is synthesized. It is introduced to a bacterial culture. The bacteria die or they do not. If they die, the molecule works. If they do not, the molecule does not work. The chemist's institutional prestige, publication record, and funding history have no influence on whether the molecule binds.

A semiconductor engineer designs a transistor at a three-nanometer process node. The transistor switches or it does not. It switches at the specified speed or it does not. It holds its state for the specified duration or it does not. There is no debate. There is no interpretation. There is measurement, and there is function.

Every person reading this paper trusts these outcomes implicitly. Nobody debates whether bridges hold. Nobody argues about whether antibiotics kill bacteria. Nobody questions whether their phone's processor computes. These things work, and the fact that they work is self-evident to everyone who uses them.

Ask a person on the street what produced these outcomes, and they will say: science.

This is where the confusion begins.

---

2. The Common Assumption

The dominant cultural model in 2026 treats science and engineering as two phases of a single enterprise. Science discovers the truth about nature. Engineering applies that truth to build useful things. Discovery is the noble phase. Application is the practical phase. In this model, science is upstream and engineering is downstream. Science leads. Engineering follows.

This model assigns status accordingly. The scientist is an explorer, a seeker of truth, a person who pushes the boundary of human knowledge. The engineer is a builder, an applier, a person who takes what the scientist found and makes it useful. In university hierarchies, in funding allocation, in cultural prestige, in media representation, the scientist outranks the engineer. The Nobel Prize is for science. The public intellectual is a scientist. The magazine cover features a physicist, not a structural engineer.

This status assignment is so deeply embedded that questioning it feels like questioning the value of knowledge itself. But the model that produces the status assignment has a structural problem: it does not describe what actually happened.

---

3. The Migration

Consider electricity. In the eighteenth century, researchers rubbed amber and observed sparks. They flew kites in storms. They built Leyden jars and shocked themselves at parties. Electricity was a scientific curiosity — a natural phenomenon being explored for understanding. There was no application. There was no engineering. There was observation, experimentation, and gradually accumulating knowledge about what electricity was and how it behaved.

Then the knowledge became reliable enough to build on. Faraday's work on electromagnetic induction was scientific discovery. The dynamo that followed was engineering. Maxwell's equations were mathematical physics. The telegraph, the motor, the power grid — these were engineering. The science phase was temporary. It produced understanding. That understanding migrated into engineering, and engineering is where it has lived ever since. Nobody calls the power grid science. Nobody calls a motor a scientific instrument. The science is finished. The engineering continues.

The same pattern repeats across every major domain.

Chemistry began as natural philosophy — alchemists seeking transformation, early chemists characterizing elements, Lavoisier establishing conservation of mass. The discoveries migrated. Chemical engineering now produces materials, fuels, pharmaceuticals, and industrial processes. The science phase identified the reactions. The engineering phase builds with them.

Biology at the molecular level followed the same path. Watson and Crick's characterization of DNA structure was scientific discovery. PCR, gene sequencing, CRISPR — these are genetic engineering. The science produced the map. The engineering builds from the map.

Optics was scientific investigation of light behavior. It became optical engineering — lenses, fiber optics, lasers, lithography systems. Thermodynamics was scientific investigation of heat and energy. It became mechanical and thermal engineering — engines, refrigeration, power systems.

In every case, the pattern is the same. A period of scientific investigation produces understanding of a natural phenomenon. The understanding reaches a threshold of reliability. At that threshold, the knowledge becomes buildable — it can be used to produce functional output that must work. At that moment, it migrates from science to engineering. The migration is one-directional. Nothing migrates back. Once a domain of knowledge is reliable enough to build on, it becomes engineering permanently, and the science phase is retrospectively understood as the developmental period that preceded the functional phase.

This is not a controversial observation. Every reader can verify it against their own experience. The things that work in daily life — transportation, communication, medicine, computation, materials, construction — are all engineering. They were all preceded by a science phase. The science phase ended when the engineering phase began.

The question this paper asks is: what does this migration leave behind?

---

4. What Remained

After the productive discoveries migrated into engineering, the word science did not become empty. Fields continued to operate under the label. Departments remained open. Journals continued publishing. Funding continued flowing. Careers continued being built.

But the fields that remained under the label science are not all doing the same thing. Two very different kinds of activity share the word, and the difference between them matters.

The first kind is formal. Mathematics proves theorems. Logic establishes valid inference. Theoretical computer science determines what is computable and at what cost. These disciplines produce conclusions that are true by internal necessity — they follow from axioms through rules of inference to results that cannot be otherwise within the system. A mathematical proof does not require experimental validation. It does not need to be replicated in a laboratory. It is true because the logic is valid, and it remains true regardless of what the physical universe does.

Formal disciplines provide tools that engineering uses — calculus for structural analysis, statistics for quality control, information theory for communication systems, computational complexity for algorithm design. But the formal disciplines themselves do not migrate into engineering because they were never empirical in the first place. They did not begin as observations of nature. They began as constructions of logical systems. They remain valid in their own domain permanently, and their validity does not depend on functional demonstration.

The second kind is different. Cosmology constructs models of the universe's origin, structure, and fate. Paleontology reconstructs ancient organisms from fragmentary physical remains. Climate science models the behavior of a planetary system with more interacting variables than any model can fully capture. Nutrition epidemiology draws causal conclusions from observational studies that cannot control for confounding variables. Psychiatry categorizes mental conditions by symptom clusters and treats them with pharmacological agents whose mechanisms of action in the relevant conditions are incompletely understood. String theory develops mathematical frameworks that are internally consistent but make no testable predictions.

These fields produce explanations. They construct narratives about how things are, were, or will be. The narratives may be interesting, intellectually rigorous, and even partially correct. But they share a defining characteristic: they cannot be turned into functional output that must work. There is no cosmological machine that must function or fail. There is no paleontological device that reality tests every day. There is no nutritional engineering in which a dietary recommendation must work the way a bridge must hold, with immediate and undeniable feedback if it does not.

These two kinds of activity — formal truth and explanatory narrative — are both called science. They are not the same thing. And neither of them is the thing that produced the bridge, the antibiotic, or the transistor. That thing migrated into engineering and is no longer called science at all.

---

5. Three Categories

The situation in 2026 is that one word covers three distinct activities with different accountability structures, different relationships to evidence, and different connections to physical reality.

Category One: Engineering. Produces functional output that must work. Includes everything that migrated from science once it became reliable enough to build on. The defining feature is functional accountability — reality provides immediate feedback that no institutional authority can override. The bridge holds or it does not. The drug binds or it does not. The chip computes or it does not.

Category Two: Formal Science. Produces logically valid conclusions within axiomatic systems. Includes mathematics, formal logic, and theoretical computer science. The defining feature is logical necessity — conclusions follow from axioms through valid inference and are true regardless of empirical observation. Formal science is legitimate, valuable, and not what people mean when they say science in ordinary conversation.

Category Three: Explanatory Narrative. Produces explanations of phenomena that cannot be turned into functional output from their current position. Includes cosmology, paleontology, portions of climate science, nutrition epidemiology, much of psychiatry, archaeology, string theory, and evolutionary biology as currently practiced. The defining feature is the absence of functional accountability — the narrative is evaluated by institutional consensus rather than by whether it produces something that must work.

These three categories are currently filed under one word. This paper argues that the mislabeling is not merely imprecise but actively harmful.

---

6. The Borrowed Authority

When all three categories share the word science, the categories with weaker accountability borrow credibility from the category with the strongest accountability — and that strongest category is no longer even called science.

The credibility that the word science carries was earned by the discoveries that became engineering. The public trusts science because bridges hold, because antibiotics work, because phones compute. This trust was built by functional demonstration over centuries. Every time a scientific discovery became reliable enough to build on and the built thing worked, the word science accumulated another increment of public trust. The trust is real and it was earned.

But the trust was earned by work that has since migrated out of the category. The things that built the trust are now called engineering. The word science retains the trust without retaining the work that generated it. This creates an arbitrage opportunity: any field that calls itself science can spend the credibility that engineering accumulated, without being subject to engineering's accountability.

When a cosmologist says "the science shows that dark matter constitutes 27% of the universe," the listener hears the word science and imports the credibility of bridges and antibiotics. The listener does not naturally distinguish between a claim backed by functional demonstration and a claim backed by a mathematical model that postulates an undetectable substance. Both are science. The word is the same. The credibility transfer is automatic.

When a nutrition scientist says "the science shows that dietary fat causes heart disease," the listener again imports the credibility of functional demonstration. The listener does not naturally distinguish between a drug that demonstrably kills bacteria in a controlled experiment and a dietary recommendation drawn from a food frequency questionnaire filled out from memory by study participants whose other variables — exercise, genetics, smoking, stress — could not be controlled.

The mislabeling is not a philosophical nicety. It produces real-world harm. The food pyramid — presented as settled science with the full authority of the word — directed the dietary behavior of hundreds of millions of people for decades. The recommendations were wrong. Dietary fat was not the primary driver of heart disease. Sugar and refined carbohydrates, which the pyramid encouraged as the dietary base, contributed to an epidemic of obesity and metabolic disease. The harm was diffuse, delayed, and enormous.

The chemical imbalance theory of depression — presented as established neuroscience — justified the prescription of billions of doses of selective serotonin reuptake inhibitors. The theory has since been largely abandoned. The field that prescribed the medication on the basis of a theory it now acknowledges was incomplete or incorrect has not recalibrated its authority claims accordingly. The word science still attaches to psychiatric diagnosis and treatment, carrying the same credibility as a pharmaceutical molecule whose mechanism is fully characterized and whose binding affinity is measured to multiple decimal places.

In each case, the harm followed the same structure: a category three claim was presented with category one authority, the public accepted it because the word was the same, and the absence of functional accountability meant the error persisted for decades before correction.

---

7. The Accountability Gap

The distinction between categories is ultimately about feedback.

Engineering has a feedback loop that is tight, immediate, and undeniable. Design, build, test. If the test fails, the design is wrong. The feedback is provided by reality itself — gravity, chemistry, electromagnetism, thermodynamics. No institutional process mediates the feedback. No committee decides whether the bridge held. The bridge held or it did not. The feedback loop is closed by physics, not by people.

This feedback loop is the source of engineering's reliability. It is not that engineers are smarter than scientists, more honest than scientists, or more rigorous than scientists. It is that engineering operates under a constraint that enforces correction: the thing must work. This constraint is external, implacable, and indifferent to institutional politics. It cannot be lobbied. It does not respond to consensus. It does not care about careers, funding, prestige, or publication records.

Category three has no equivalent constraint. If a cosmological model is wrong, nothing collapses. No machine stops working. No patient dies from an incorrect model of dark matter. The model persists or is revised based on institutional processes — peer review, replication attempts, consensus conferences, funding decisions. These processes are conducted by people, and people are subject to incentive structures that physics is not.

The incentive structures of academic science reward novelty, confidence, and narrative coherence. A paper that says "we found a bone and we do not know what it means" is not publishable. A paper that says "we found a bone and it represents a new species that fills a critical gap in the evolutionary lineage" is publishable, fundable, and career-making. The incentive is to construct the fullest narrative the evidence can be made to support, not to state the most conservative conclusion the evidence actually warrants.

Peer review — the institutional mechanism that is supposed to provide accountability — operates within the same incentive structure. The reviewers are members of the same community, competing for the same funding, publishing in the same journals, building careers on the same narrative frameworks. A reviewer who challenges the fundamental assumptions of a field is challenging the foundation of his own career. The system selects for reviewers who share the field's assumptions, which means peer review validates work against the community's existing commitments rather than against external reality.

This is not a claim that scientists are corrupt. It is a structural observation. The incentive structure of category three research does not include a mechanism equivalent to the bridge collapsing. The correction mechanism is social rather than physical, slow rather than immediate, and mediated by the same community that produced the claim rather than by an external reality that is indifferent to it.

---

8. The Blocking Problem

Category three does not only borrow authority from category one. It actively directs category one's work.

This is the most consequential harm of the mislabeling, and it operates through the institutional structures that allocate funding, set research priorities, and determine which questions are considered important.

When a category three field establishes a narrative, that narrative generates research questions. The research questions generate grant proposals. The grant proposals direct experimental work in category one fields. The flow of influence is from narrative to experiment — from the field that cannot build anything to the field that can.

Cosmological models predict that dark matter particles should exist. Particle physics experiments are designed, funded, and built to detect them. Detectors run for years. The particles are not found. The response is not to question the cosmological model. The response is to build more sensitive detectors, to propose new categories of dark matter particle, to extend the search. The narrative survives the null result and generates the next round of experimental work. The category one field — particle physics, which measures real phenomena to twelve decimal places of precision — is being directed by the category three field — cosmology, which postulates undetectable entities to preserve a mathematical model.

This redirection has an opportunity cost that is concrete and measurable. Every dollar, every researcher, every detector, every year of experimental effort directed toward finding dark matter is a dollar, a researcher, a detector, and a year not directed toward questions that particle physics could answer on its own terms — questions about the discrete structure of matter, about the integer foundations of quantum mechanics, about the phenomena that particle physics has already demonstrated it can measure with extraordinary precision.

A specific case study illustrates the depth of this problem.

8.1 The Integer Problem

Quantum electrodynamics is the most precisely confirmed theory in the history of physics. Its predictions of the electron's anomalous magnetic moment agree with experimental measurement to twelve decimal places. No other theory in any field of science has achieved this level of confirmation.

QED operates on integers. Particles come in integer counts. Interactions occur through discrete events. Quantum numbers are integers or simple fractions of integers. The photon is one photon. Two photons is two photons. There is no half-photon. The discrete structure is not a mathematical convenience or an approximation. It is what the experiments measure, and they measure it to twelve digits of precision.

General relativity operates on real numbers. Spacetime is modeled as a smooth continuous manifold. Curvature varies continuously. The mathematics requires real-valued tensors over continuous coordinate systems. The predictions of general relativity are confirmed — gravitational lensing, frame dragging, gravitational waves — but to far fewer digits of precision than QED's predictions.

When physicists attempt to combine the two theories — quantum field theory on curved spacetime — the result is the vacuum energy catastrophe. The naive calculation of vacuum energy density from quantum field theory produces a value that exceeds the observed value by a factor of approximately 10^122. This is not a small discrepancy. It is 122 orders of magnitude. It is the worst quantitative prediction in the history of physics.

The standard response is renormalization — a set of mathematical procedures that remove infinities from calculations by subtracting them in a controlled way. Renormalization produces correct numerical answers. The predictions that emerge after renormalization match experiment. But the procedure itself is formally non-physical. It takes an infinite result — a result that is physically meaningless — performs a subtraction that has no physical interpretation, and keeps the finite remainder. The finite remainder agrees with measurement, but the procedure that produced it involves steps that do not correspond to anything that happens in the physical universe.

The situation in 2026 is this: QED demonstrates to twelve decimal places that the universe operates on discrete integer quantities at the fundamental level. General relativity demonstrates to fewer decimal places that gravity is well-described by continuous mathematics at large scales. When the two are combined, the continuous mathematics produces nonsense — 10^122 off — and the nonsense is rescued by formally non-physical procedures.

The institutional response has been to continue attempting reconciliation within the continuous mathematical framework. String theory, loop quantum gravity, and other unification programs all attempt to derive discrete quantum behavior from fundamentally continuous mathematical structures. Decades of effort in this direction have produced no testable predictions.

The alternative — beginning from QED's demonstrated integer structure and working upward to ask what gravity looks like when the underlying mathematics is discrete rather than continuous — receives minimal institutional support. Not because it has been tried and failed, but because the continuous framework arrived first, carries the prestige of general relativity's confirmed predictions, and occupies the institutional high ground from which research priorities are set.

The blocking is precise: the less-confirmed theory's mathematical assumptions are treated as the foundation that the more-confirmed theory must conform to. The twelve-digit integer theory is being forced to accommodate the four-digit continuous theory, rather than the reverse. Category three — cosmology, which uses continuous mathematics to construct narrative models of the universe — is directing category one — particle physics, which uses discrete mathematics to make the most precise measurements in science.

---

9. The Migration Test

This paper offers a tool for evaluating any claim currently operating under the label science.

Step one. Has this claim been turned into engineering? Has it produced functional output that must work — a device, a material, a process, a treatment whose failure would be immediate and undeniable? If yes, the claim has migrated into engineering. The science phase is complete. The claim is validated not by the word science but by the thing that works.

Step two. If the claim has not migrated, ask: could it? Is there a path from the current claim to functional output? Not a theoretical path in which further research might eventually lead somewhere. A concrete path in which the claim, as currently stated, could produce something that must work.

Step three. If the claim cannot produce functional output because it is unfalsifiable, because the evidence cannot be controlled or reproduced, because the phenomenon cannot be isolated from confounding variables, or because the claim is about events in the unreachable past or the unobservable distance, then the claim is category three regardless of its institutional prestige, its publication history, or the number of experts who endorse it.

This is not a test of whether a claim is interesting, intellectually stimulating, or potentially useful as a hypothesis-generating framework. Category three claims can be all of those things. The test determines what kind of epistemic authority the claim is entitled to. A category three claim is entitled to the authority of a plausible explanation. It is not entitled to the authority of a functional demonstration. The word science, as currently used, does not distinguish between these two very different levels of authority.

Examples of the test in application:

"This antibiotic kills this bacterium at this concentration." Apply the test. Has it been turned into engineering? Yes. The antibiotic is manufactured and prescribed. The functional output is the dead bacterium. Category one. Full epistemic authority.

"The Pythagorean theorem states that in a right triangle, the square of the hypotenuse equals the sum of the squares of the other two sides." Apply the test. This is not an empirical claim. It is a logical necessity within Euclidean geometry. Category two. Full epistemic authority within its axiomatic system.

“Dark matter constitutes 27% of the universe’s mass-energy content.” Apply the test. Has it been turned into engineering? No. Could it be? Not from its current position — dark matter has never been detected by any instrument. The claim exists to preserve a mathematical model that does not match observation without it. Category three. Entitled to the authority of a plausible explanation within a specific mathematical framework. Not entitled to the authority of a functional demonstration.

“This fossil represents a transitional species between terrestrial mammals and modern cetaceans.” Apply the test. Has it been turned into engineering? No. Could it be? Not from its current position — the claim is about an unreachable past event reconstructed from fragmentary remains. The reconstruction cannot be functionally validated because the animal cannot be built and tested. Whether the reconstructed organism could actually function — manage its body heat, sustain its cardiovascular requirements, acquire sufficient calories, survive in its proposed environment — is not part of the evaluation process that produced the claim. Category three.

“Dietary fat consumption causes cardiovascular disease.” Apply the test. Has it been turned into engineering? It was treated as though it had — dietary guidelines were issued, food products were reformulated, public health policy was built on it. But the functional output — healthier populations — did not materialize. The populations that followed the guidelines did not become healthier. They became sicker. The claim was treated as category one and implemented as engineering, but it was category three — an explanatory narrative drawn from uncontrolled observational studies. The implementation failed because the accountability gap allowed the claim to reach engineering-level influence without passing engineering-level validation.

---

10. The Formal Science Exception

This paper does not claim that everything must become engineering to be valid.

Formal science — mathematics, logic, theoretical computer science — occupies a category of its own. Its validity does not depend on functional demonstration because its claims are not empirical. A mathematical proof is not an observation about the physical world. It is a demonstration that a conclusion follows necessarily from stated axioms through valid inference rules. The proof is true within its system regardless of what happens in the physical universe.

Formal science provides tools that engineering uses. Calculus enables structural analysis. Statistics enables quality control. Information theory enables communication system design. Computational complexity theory enables algorithm engineering. The tools are indispensable. But the formal disciplines that produce them do not migrate into engineering the way empirical discoveries do, because they were never empirical. They began as logical constructions, they remain logical constructions, and their validity is permanent within their axiomatic systems.

The distinction matters because it prevents a misreading of this paper’s argument. The paper does not claim that the migration test is the only measure of validity. Formal science is valid without passing the migration test. The migration test applies to empirical claims — claims about the physical world — and determines whether those claims have achieved functional accountability or remain at the level of explanatory narrative.

A further subtlety is worth stating. Formal science can be applied incorrectly to empirical domains. A mathematical framework can be internally consistent — every derivation valid, every proof sound — while describing nothing real. String theory is a potential example: the mathematics may be flawless while the physical content is zero. Internal validity is necessary for formal science. It is not sufficient for empirical relevance. When formal science is used to support empirical claims, the claims must be evaluated on their empirical merits, not on the validity of the mathematics. Valid mathematics applied to wrong assumptions produces precise nonsense.

---

11. The Field Map

Where does each major field sit in 2026? Some fields occupy a single category. Others split, with portions having migrated to engineering while other portions remain explanatory narrative.

Physics is split. Particle physics at the experimental level is category one engineering — detectors are built, measurements are made, the apparatus either works or it does not, and the results are confirmed to extraordinary precision. Classical mechanics, electromagnetism, and thermodynamics completed their migration to engineering centuries ago. Cosmology is category three — narrative models of the universe's origin, structure, and fate, evaluated by mathematical consistency and institutional consensus rather than by functional output. String theory is category three — internally consistent mathematics that has produced no testable prediction in decades of development.

Chemistry has largely migrated. Synthetic chemistry, materials chemistry, and biochemistry at the molecular level are engineering. Chemical processes either produce the target molecule or they do not. The remaining science-labeled portions — theoretical chemistry, computational chemistry at the exploratory level — are either formal science (mathematical modeling of molecular behavior) or hypothesis generation awaiting migration.

Biology is split. Molecular biology, genetics, and microbiology at the applied level have migrated to bioengineering and pharmaceutical engineering. Evolutionary biology as currently practiced is category three — a narrative framework built on incomplete evidence that cannot produce functional output. Ecology is largely category three — observational, model-dependent, and unable to produce controlled functional tests of its major claims.

Medicine is deeply split. Surgery is engineering — the procedure works or the patient does not recover. Pharmaceutical development at the molecular level is engineering — the molecule binds or it does not. Emergency medicine is engineering — the intervention stabilizes the patient or it does not. Psychiatry is largely category three — diagnosis by symptom checklist, treatment by pharmacological trial and error, theoretical frameworks that have been abandoned without corresponding reduction in prescribing authority. Nutrition science is largely category three, with a documented history of category three claims being implemented as though they were category one and causing population-scale harm.

Earth sciences are split. Geology at the applied level — mining, petroleum engineering, geotechnical engineering — has migrated. Seismology is partially engineering, partially predictive narrative. Climate science is largely category three — models of a system too complex to control or fully characterize, with predictions evaluated over timescales too long for the feedback loop to close within a career or a funding cycle. Paleontology is category three — narrative reconstruction from fragments without engineering validation of whether the reconstructed organisms could function as living systems.

Mathematics is category two entirely. Formal, axiomatic, valid by logical necessity.

Computer science splits. Applied computer science — software engineering, hardware design, networking — is category one engineering. Theoretical computer science — computability theory, complexity theory, formal verification — is category two formal science. Artificial intelligence research sits uneasily across categories, with applied machine learning being engineering (the model performs or it does not) and theoretical AI alignment being closer to category three (narrative frameworks about future systems that do not yet exist).

This map is offered as a starting point. Readers will disagree with specific placements. The value of the map is not in any individual placement but in the existence of the categories themselves. Once the three categories are visible, every reader can apply the migration test to any field and determine for themselves where it sits. The map is an invitation to perform the test, not a final verdict.

---

12. What This Paper Does Not Claim

Category three is not worthless. Explanatory narrative serves genuine functions. It generates hypotheses that category one can test. It provides frameworks that organize observations into patterns that might eventually become engineering-ready. It satisfies the human desire to understand the world, which is a legitimate intellectual motivation even when the understanding cannot be functionally validated.

This paper does not claim that all category three work should stop. It does not claim that cosmologists, paleontologists, climate scientists, or psychiatrists are fraudulent. It does not claim that narrative has no place in intellectual life.

This paper claims three things.

First, that the three categories exist and are structurally different in their relationship to evidence, accountability, and physical reality.

Second, that housing all three categories under one word allows the category with the weakest accountability to borrow authority from the category with the strongest accountability, and that this borrowing produces real-world harm when category three claims are implemented with category one confidence.

Third, that category three should not direct category one's research priorities, because when explanatory narrative steers functional research, the functional research is diverted from questions it could answer toward questions generated by narratives that may be wrong, and the opportunity cost of this diversion is measured in decades of misdirected effort.

The fix is not to destroy category three. The fix is honest labeling. When someone says "the science says," the listener should be able to determine which category is speaking. A functional demonstration and an explanatory narrative are not the same thing. They should not be presented as the same thing. And the word science, as currently used, presents them as the same thing.

---

13. Conclusion

The division described in this paper already happened. The productive discoveries migrated into engineering. The formal disciplines remained valid in their own domain. The explanatory narratives remained under the label science, carrying the authority that the migrated discoveries earned.

This is not a crisis to be solved. It is a situation to be named. The naming itself is most of the work, because once the three categories are visible, the structural problems — borrowed authority, absent accountability, misdirected research — become visible too.

The bridge holds because engineering requires it to hold. The theorem is true because logic requires it to be true. The narrative persists because the institution permits it to persist. These are three different relationships to reality, operating under three different accountability structures, producing three different kinds of confidence.

One word for all three is one word too few.

END HOWL-CULT-11-2026

Registry: HOWL-CULT-11-2026

Status: Position paper

Claims: Three-category taxonomy of activities currently labeled science; structural analysis of credibility transfer between categories; identification of research priority misdirection from explanatory narrative to functional research

Boundary: Does not claim category three is worthless; does not claim all work must become engineering; claims honest labeling and appropriate authority assignment

[**HOWL-CULT-11-2026**] Engineering and Science. Github: https://github.com/ghowland/papers/tree/main/papers/CULT
CULT-11-2026